

Loyola Marymount University

CMSI 3300 Artificial Intelligence

Fall 2024

Professor Lanyu Shang

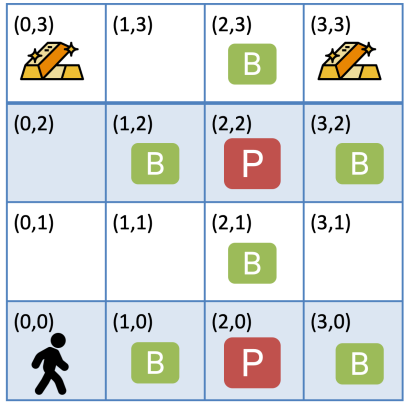
Announcements

- **Recitation**: Tuesdays @6 pm in Keck Annex
- **Groups:** information is available on Brightspace
  - If you have any questions, please let me know asap.
- Due **this Friday, 9/6 @5 pm**:
  - Classwork 1 (individual-based)
- Upcoming:
  - Classwork 2 (group-based) will be released this Thursday

This Lecture

- Propositional Logic & Truth Table
- Knowledge Base Structure
- Model Checking

Last Lecture



Knowledge Base

Rule →
- If a cell has a breeze, then a pit is adjacent, vice versa.

Facts {
- Cell (0,0) has no breeze.
- Cell (1, 0) has a breeze.
- Cell (0, 1) has no breeze.

Inference Engine → Answer

Query

Is Cell (1,1) a pit?

Propositional Logic
- **Syntax**
- **Semantics**

Propositional Logic

- Propositions
  - **Boolean** variable (**`T`**/**`F`**)
  - Make **assertions** about aspects of the reasoning **environment**
  - E.g., $B_{1,0}$: There is a breeze in (1,0)
- Worlds
  - A possible **instantiation** of all propositions (i.e., an assignment of **`True`**, **`False`** to all propositions)
  - E.g., one row in a truth table

Propositional Syntax

- Atomic Sentence (e.g., $B_{1,0}$, $\neg P_{1,1}$)
- Complex Sentence
  - Logical grouping **()** and **connectives**
  - **5 Connectives** ($\neg$, $\wedge$, $\vee$, $\Rightarrow$, $\Leftrightarrow$)

Truth Table

World	P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$	$P \Rightarrow Q$	$P \Leftrightarrow Q$
w_1	F	F	T	F	F	T	T
w_2	F	T	T	F	T	T	F
w_3	T	F	F	F	T	F	F
w_4	T	T	F	T	T	T	T

Logical Equivalence

- Two sentences are **logically equivalent** (≡) if they have the same truth-table values for each world.
  - $P \Rightarrow Q \equiv \neg P \vee Q$
  - $P \Leftrightarrow Q \equiv (P \Rightarrow Q) \wedge (Q \Rightarrow P)$

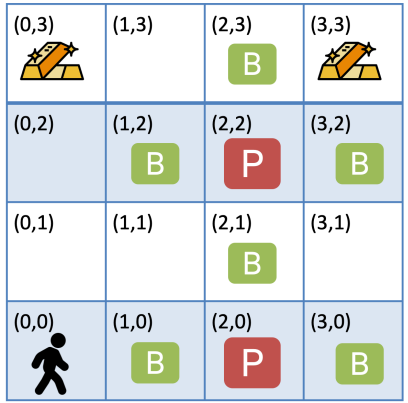
Truth Table

World	P	Q	$\neg P$	$\neg P \vee Q$	$P \Rightarrow Q$	$Q \Rightarrow P$	$(P \Rightarrow Q) \wedge (Q \Rightarrow P)$	$P \Leftrightarrow Q$
w_1	F	F	T	T	T	T	T	T
w_2	F	T	T	T	T	F	F	F
w_3	T	F	F	F	F	T	F	F
w_4	T	T	F	T	T	T	T	T

Next…

KNOWLEDGE BASE STRUCTURE

Knowledge Base Structure



Feedback loop

Knowledge Base

Online Learning/ Experience

Fell in a pit at Cell (2,0)

- If a cell has a breeze, then a pit is adjacent, vice versa.
- Cell (0,0) has no breeze.
- Cell (1, 0) has a breeze.
- Cell (0, 1) has no breeze.
- Cell (2,0) has a pit.

Inference Engine

Answer

Propositional Logic

- **Syntax**
- **Semantics**

Query

Is Cell (1,1) a pit?

Knowledge Base Structures

- A collection of **logical sentences** that we hold to be true about the environment.
- Our knowledge base can **grow over time** from its initial state through what is known as **online learning**, where the agent acquires experiences from the environment in which it is deployed.

Wumpus World Example

- What **propositions** (i.e., symbols / variables for T/F aspects of the environment) will we need to avoid Pits?

  $P_{x,y}$ and $B_{x,y}$

- What **rules** (i.e., propositional sentences that describe how the world works) should we include for our problem? How to phrase them?

  $B_{x,y} \Leftrightarrow P_{x-1,y} \vee P_{x+1,y} \vee P_{x-1,y} \vee P_{x+1,y} \; \forall \; x, y$

Wumpus World Example

- What **facts** (i.e., typically atomic sentences that are true in the currently observed environment) should we include for the **initial state** of our problem?

(0,3)	(1,3)	(2,3)	(3,3)
(0,2)	(1,2)	(2,2)	(3,2)
(0,1)	(1,1)	(2,1)	(3,1)
(0,0)	(1,0)	(2,0)	(3,0)



- $\neg P_{0,0}$
- $\neg B_{0,0}$

Wumpus World Example

- How can we view our KB in terms of propositional connectives?
  - Conjunctions (a giant complex sentence)

$$KB = Rule1 \wedge Rule2 \wedge \cdots \wedge Fact1 \wedge Fact2 \wedge \cdots$$

- What are the benefit?
  - Easy to update: KB.*tell*(new_fact): $KB = KB \wedge new_fact$
  - Commutative: $Rule1 \wedge Rule2 \wedge \cdots = Rule2 \wedge Rule1 \wedge \cdots$
  - Convenient to enumerate (like a set)

Next…

MODEL CHECKING

Propositional Inference

- The process of determining the truth of certain **sentences** that are **not stated explicitly** in a KB.
  - Memorize vs. inference
  - Deduce something that we knew, but wasn't obvious
- A propositional sentence is often denoted using Greek letters (e.g., α, β).

"Model" – Interpreting Inference

- The **model** of a propositional sentence α is the **set** of all **worlds** $\{w_1, w_2, \dots\}$ in which α is **true**. Formally,

$$M(\alpha) = \{w_1, w_2, \dots\} \text{ where } \alpha \text{ is true}$$

- Example

World	P	Q	$\alpha = \neg Q$
w_1	F	F	T ✓
w_2	F	T	F
w_3	T	F	T ✓
w_4	T	T	F

Worlds

Sentences

What is $M(\alpha)$?

$M(\alpha) = \{w_1, w_3\}$

Model of a KB

- Given that

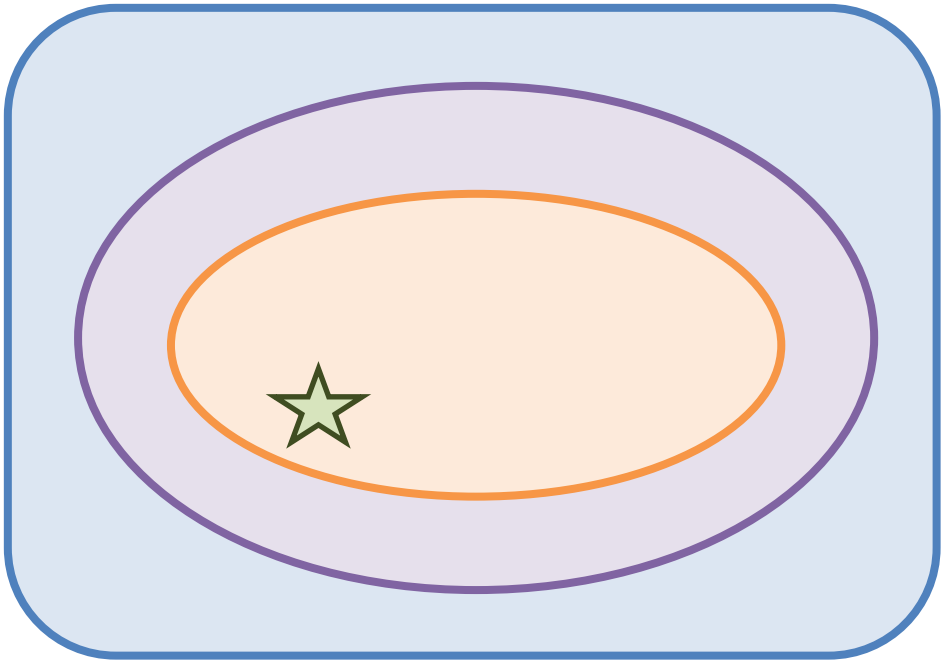
$$KB = Rule1 \wedge Rule2 \wedge \cdots \wedge Fact1 \wedge Fact2 \wedge \cdots$$

- Property of M(KB):

$$M(KB \wedge newFact) \boxed{\subseteq} M(KB)$$

$\subset \supset \subseteq \supseteq$

Model of a KB



M(KB)

M(KB ∧ new_fact)

All Worlds

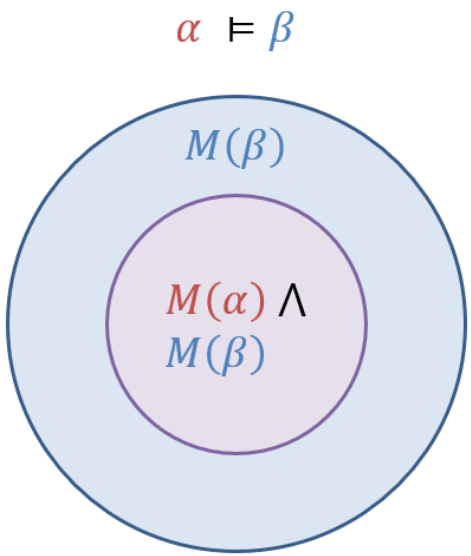
True State

Wumpus World Example

- Consider that we wish to query our KB to **determine if a Pit exists at (2,0)**, i.e., we want to know the truth value of $P_{2,0}$ in M(KB).
- There are 3 possible answers to this query.
  - Yes! $P_{2,0}\ \forall\ m \in M(KB)$
  - No! $\neg P_{2,0}\ \forall\ m \in M(KB)$
  - Maybe? $\exists\ m \in M(KB)\ s.t.\ P_{2,0}$ and $\exists\ m \in M(KB)\ s.t.\ \neg P_{2,0}$

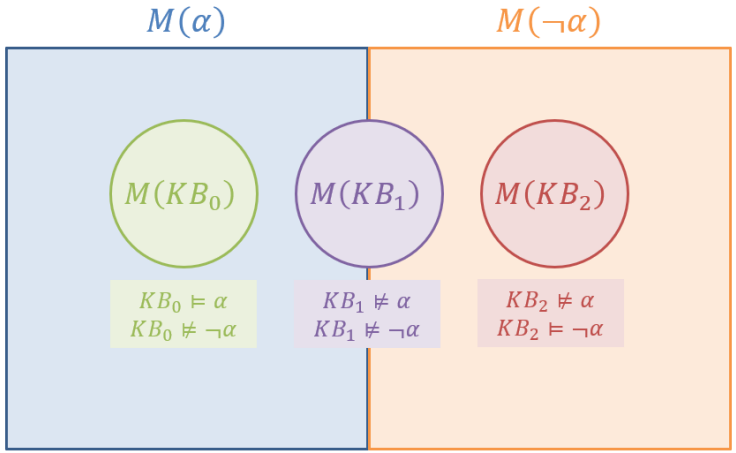
Entailment

- For two propositional sentences α, β,
  - α **entails** β means β follows from α
  - $\alpha \models \beta$ if the models of α are a **subset** of the models of β
  - Formally, $\alpha \models \beta \Leftrightarrow M(\alpha) \subseteq M(\beta)$



Wumpus World Example

- Query: if a Pit exists at (2,0)
  - Yes
  - No
  - Maybe



Model Checking

- To determine if you have enough information to make a conclusion
- Example: to determine the state of entailment for some sentence α in our KB, you'll need 2 tests
  - $KB \models \alpha$ and
  - $KB \models \neg\alpha$

Example

- 3 propositions: R, P, S
- $KB = (\neg P \vee S) \wedge \neg R$

Example

World	R	P	S	$KB = (\neg P \vee S) \wedge \neg R$	$\neg R$	$\neg P$
w_0	F	F	F	T	T	T
w_1	F	F	T	T	T	T
w_2	F	T	F	F	T	F
w_3	F	T	T	T	T	F
w_4	T	F	F	F	F	T
w_5	T	F	T	F	F	T
w_6	T	T	F	F	F	F
w_7	T	T	T	F	F	F

$M(KB)$
$M(R)$
$M(P)$
$M(\neg R)$
$M(\neg P)$

$KB \models R?$ No
$KB \models \neg R?$ Yes
$KB \models P?$ No
$KB \models \neg P?$ No